

MTHE 351 – Fall 2026

Practice Problem Set 0: Elementary Set Theory

1. An insurance company classifies its set S of policyholders by using the following sets:

$$\begin{aligned}A &= \{x \in S : x \text{ drives a subcompact car}\}, \\B &= \{x \in S : x \text{ drives a car that is more than 5 years old}\} \\C &= \{x \in S : x \text{ is married}\}, \\D &= \{x \in S : x \text{ is over 20 years of age}\}, \\E &= \{x \in S : x \text{ is male}\}.\end{aligned}$$

Express each of the following subsets of S in terms of A , B , C , D and E .

- (a) Female policyholders over 20 years of age.
 - (b) Policyholders who are male or drive cars more than 5 years old.
 - (c) Female policyholders over 20 years of age who drive subcompact cars.
 - (d) Male policyholders who are either married or over 20 years of age and do not drive subcompact cars.
2. Let $\mathbb{N}$ and $\mathbb{R}$ denote the set of positive integers and the set of real numbers, respectively. Define the following sets:

$$\begin{aligned}E &= \{x : x = 2n \text{ for } n \in \mathbb{N}\}, \\F &= \{x : x = 2n - 1 \text{ for } n \in \mathbb{N}\}, \\A &= \{x \in \mathbb{R} : -4 < x < 3\}, \\B &= \{x \in \mathbb{R} : -1 < x < 7\}, \\C &= \{x \in \mathbb{R} : x^2 = -2\}.\end{aligned}$$

- (a) Describe the sets E , F , and C in words.
 - (b) Find $A \cup B$, AB , $A - B$, and A^c .
 - (c) Find $F \cup E$, FC , and C^c .
3. Let $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ and $B = \{z \in \mathbb{R} : 0 \leq z \leq 1\} = [0, 1]$. Describe the following sets in words: $A \times B$, $A \times \{0\}$, and $A \times \{1/2\}$.
4. Prove De Morgan's law $(A \cup B)^c = A^c B^c$ by showing that $(A \cup B)^c \subset A^c B^c$ and $A^c B^c \subset (A \cup B)^c$.

5. Show that for any sets A and B in a universal set S ,

$$A - (A - B) = AB$$

by using elementary properties of set operations (such as De Morgan's law, distributivity, etc).

6. Let $A = \{x, y\}$, $B = \{0, 1\}$, and $C = \{-1, 0, 1\}$. Find the following sets:

(a) $A \times B$.

(b) $A \times B \times C$.

7. For $n = 1, 2, \dots$ consider the following intervals of the real line: $A_n = [0, 1 + 1/n)$, $B_n = [1, 1 + 1/n)$, $C_n = (1, 1 + 1/n)$, $D_n = [0, 1 - 1/n]$.

(a) Determine $\bigcap_{n=1}^{\infty} A_n$, $\bigcap_{n=1}^{\infty} B_n$, and $\bigcap_{n=1}^{\infty} C_n$.

(b) Determine $\bigcup_{n=1}^{\infty} D_n$.